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**AIML B2**
**Fuzzy Logic**

**Exp. 4 – Plotting Fuzzy C-Means clustering**

**Code :-**

```
% Lab 4 : Fuzzy C-Means Clustering

clc; clear; close all;


% --- Step 1: Load dataset (example: 2D points) ---

data = [randn(50,2)+1; randn(50,2)+5; randn(50,2)+[8,1]]; % synthetic clusters

n = size(data,1);


% --- Step 2: Parameters ---

c = 3;          % number of clusters

m = 2;          % fuzziness parameter (m > 1)

max_iter = 100;    % maximum iterations

epsilon = 1e-5;    % stopping criterion


% --- Step 3: Initialize membership matrix randomly ---

U = rand(n,c);

U = U ./ sum(U,2);   % normalize rows so they sum to 1


% --- Step 4: FCM algorithm ---

for iter = 1:max_iter

   % Compute cluster centers

   um = U.^m;                          % fuzzy weights
```

```
    centers = (um' * data) ./ sum(um,1)';     % weighted average

    % Update membership matrix
    dist = zeros(n,c);
    for j = 1:c
        dist(:,j) = sqrt(sum((data - centers(j,:)).^2,2));
    end

    % Avoid division by zero
    dist = max(dist, eps);

    newU = zeros(n,c);
    for j = 1:c
        newU(:,j) = 1 ./ sum((dist(:,j) ./ dist).^ (2/(m-1)), 2);
    end

    % Check convergence
    if max(max(abs(newU - U))) < epsilon
        break;
    end
    U = newU;
end

% --- Step 5: Assign clusters ---
[~, cluster_idx] = max(U,[],2);

% --- Step 6: Plot results ---
figure;
```

```
gscatter(data(:,1), data(:,2), cluster_idx);

hold on;

plot(centers(:,1), centers(:,2), 'kx', 'MarkerSize', 12, 'LineWidth', 2);

title('Fuzzy C-Means Clustering');

xlabel('Feature 1'); ylabel('Feature 2');

legend('Cluster 1','Cluster 2','Cluster 3','Centers');

grid on;
```

```
% assume data is Nx2 and cluster_idx contains integers 1..k

colors = lines(max(cluster_idx));

hold on

for c = 1:max(cluster_idx)

  idx = cluster_idx==c;

  scatter(data(idx,1), data(idx,2), 36, colors(c,:), 'filled');

end

hold off
```

**Output:-**